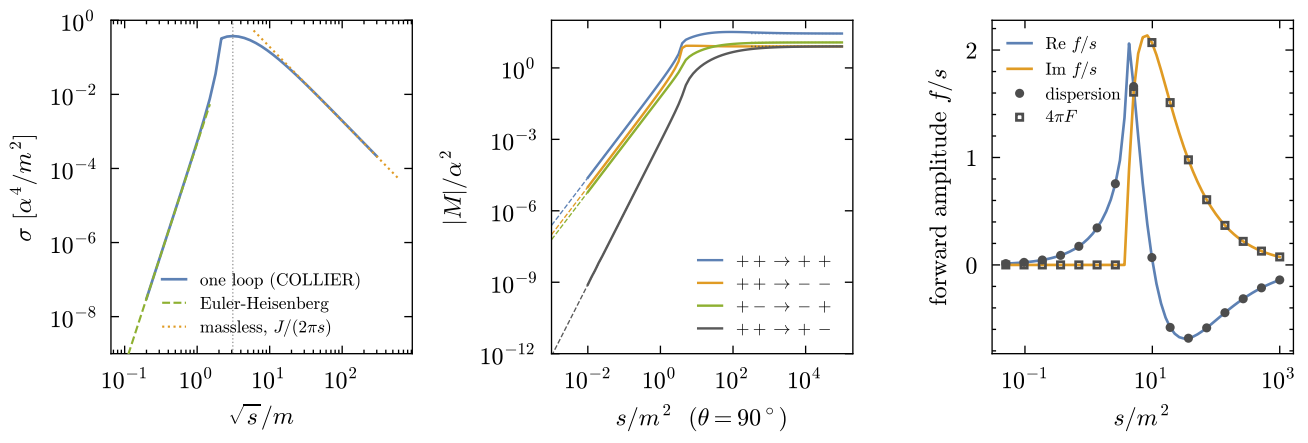
10^0
10^{-2}
10^{-4}
10^{-6}
10^{-8}
$\sigma\ [\alpha^4/m^2]$
one loop (COLLIER)
Euler-Heisenberg
massless, $J/(2\pi s)$
10^{-1}
10^0
10^1
10^2
$\sqrt{s}/m$
$|M|/\alpha^2$
10^0
10^{-3}
10^{-6}
10^{-9}
10^{-12}
$++ \to ++$
$++ \to --$
$+- \to -+$
$++ \to +-$
10^{-2}
10^0
10^2
10^4
s/m^2 $(\theta = 90^\circ)$
forward amplitude f/s
Re f/s
Im f/s
dispersion
$4\pi F$
2
1
0
10^{-1}
10^1
10^3
s/m^2